

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

*I Beffy A Shanika(192511387) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

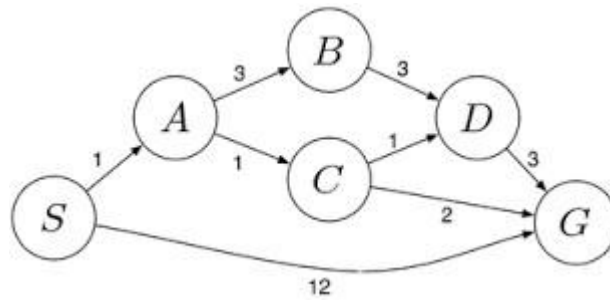
beffy

Annexure A

Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.
2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
 - i. What are the percept's for this agent?
 - ii. Characterize the operating environment.
 - iii. What are the actions the agent can take?
 - iv. How can one evaluate the performance of the agent?
 - v. What sort of agent architecture do you think is most suitable for this agent? Why?
3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.
4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.
5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse S to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

1. Water Jug Problem (4-Gallon and 3-Gallon Jugs)

Problem

Obtain exactly 2 gallons in the 4-gallon jug.

Initial State

(0,0)

where:

- First value = Water in 4-gallon jug
- Second value = Water in 3-gallon jug

Step	Action	State (4G,3G)
1	Fill 3-gallon jug	(0,3)
2	Pour 3G into 4G	(3,0)
3	Fill 3-gallon jug again	(3,3)
4	Pour from 3G into 4G until full	(4,2)
5	Empty 4-gallon jug	(0,2)
6	Pour remaining 2 gallons into 4G	(2,0)

Final State

(2,0)

Hence, the 4-gallon jug contains exactly 2 gallons.

2.Mars Rover Agent

i. Percepts

- Camera images
- Soil and rock composition
- Temperature
- Terrain information
- Obstacle detection
- Battery level
- Position (GPS/Inertial sensors)

ii. Operating Environment

- Partially Observable
- Sequential
- Dynamic
- Continuous
- Single Agent
- Unknown Environment

iii. Actions

- Move forward/backward
- Turn left/right
- Collect rock samples
- Drill soil
- Capture images
- Analyze samples
- Send data to Earth
- Recharge using solar panels

iv. Performance Measure

- Successful exploration
- Accurate sample collection
- Safe navigation
- Less energy consumption
- Maximum scientific data transmitted

v. Suitable Agent Architecture

Model-Based Goal-Based Agent

Reason:

- Maintains an internal model of Mars.
- Plans safe paths.
- Avoids obstacles.
- Achieves exploration goals efficiently.

3. 8-Queens Problem Formulation

Initial State

Empty chessboard with no queens.

State Space

All possible placements of queens on the chessboard.

Actions

Place one queen in a safe position in the next column.

Goal Test

All 8 queens are placed such that:

- No two queens share the same row.

- No two queens share the same column.
- No two queens share the same diagonal.

Path Cost

Each queen placement has cost 1.

Total solution cost = 8.

Search Strategy

- Breadth First Search (BFS)
- Depth First Search (DFS)
- Backtracking (most efficient)
- Constraint Satisfaction Problem (CSP)

4. OLA Cab Booking Problem

Type of Problem-Solving Agent

Utility-Based Agent

Reason:

The customer selects the cab (Mini, Micro, Sedan, Shared, Prime, etc.) based on comfort, fare, and travel time. The agent chooses the option that provides the highest utility.

Pseudocode

Algorithm OLA_Cab_Booking

Start

Input Source

Input Destination

Display available cab types

Mini

Micro

Sedan

Prime

Shared

For each cab

 Calculate fare

 Calculate arrival time

 Calculate comfort level

End For

Select cab with maximum utility

Book cab

Display

Cab Type

Driver Details

Estimated Fare

Estimated Arrival Time

End

5.Uniform Cost Search (UCS)

Given Graph

Edges:

- $S \rightarrow A = 1$
- $S \rightarrow G = 12$
- $A \rightarrow B = 3$
- $A \rightarrow C = 1$
- $B \rightarrow D = 3$
- $C \rightarrow D = 1$
- $C \rightarrow G = 2$
- $D \rightarrow G = 3$

UCS Steps

Step	Expanded Node	Path Cost
1	S	0
2	A	1
3	C	2
4	D	3
5	G	4

(UCS stops when **G** is removed from the priority queue.)

Cost Calculation

Possible paths:

S → G

○ Cost = **12**

S → A → B → D → G

○ Cost = $1 + 3 + 3 + 3 = \mathbf{10}$

S → A → C → D → G

○ Cost = $1 + 1 + 1 + 3 = \mathbf{6}$

S → A → C → G

○ Cost = $1 + 1 + 2 = \mathbf{4}$

UCS Expansion Order

S

↓

A

↓

C

↓

D

↓

G

Final Least-Cost Path: S → A → C → G

Total Cost: $1 + 1 + 2 = 4$

